# EXPERIMENT NO 8

**AIM:** To study the junction field effect transistor as an amplifier.

**PROBLEM STATEMENT:** To design, simulate, and analyze a Junction Field Effect Transistor (JFET) common-source amplifier using PSPICE, and to observe its time-domain (transient) response and frequency response characteristics.

**SOFTWARE REQUIRED:** Pspice Simulator

**COMPONENTS REQUIRED:**

| | |
|---|---|
| **JFET Transistor** | J2N3819 (N-Channel JFET) |
| **Drain Resistor** | 4 kΩ |
| **Source Resistor** | 1 kΩ |
| **Gate Resistor** | 3.3 MΩ |
| **Load Resistor** | 10 kΩ |
| **Input Coupling Capacitor** | 2.2 $\mu$F |
| **Output Coupling Capacitor** | 2.2 $\mu$F |
| **Bypass Capacitor** | 10 $\mu$F |
| **DC Power Supply** | 20 V DC |
| **AC Signal Source** | VSIN |

**THEORY:**

A Junction Field Effect Transistor (JFET) operated in the saturation region acts as a voltage-controlled current source. In a **Common-Source (CS)** configuration:

- **Self-Biasing Mechanism:** The gate bias resistor $R_1$ grounds the gate ($V_G \approx 0$ V). Current $I_D$ flowing through $R_2$ elevates the source voltage $V_S = I_D R_2$, producing a negative gate-to-source bias voltage $V_{GS} = -I_D R_2$ necessary for stable saturation-region operation.
- **Coupling Capacitors ($C_1, C_2$):** Block DC biasing components from interfering with the AC source and load, allowing only AC signals to pass through.

- **Bypass Capacitor ($C_3$):** Bypasses AC signals around $R_2$ to ground, preventing AC negative feedback and preserving high voltage gain.
- **Phase Inversion:** The common-source JFET amplifier provides a 180° phase shift between the input signal voltage $V_{in}$ and output signal voltage $V_{out}$.

**CIRCUIT DIAGRAM:**

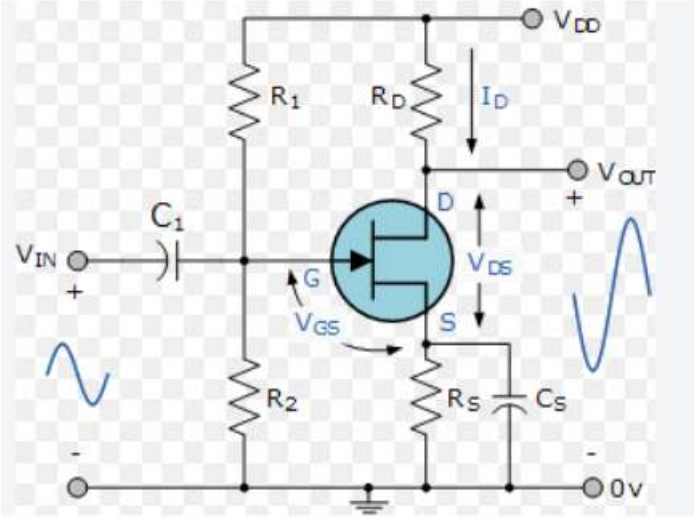



Fig: JFET as an Amplifier

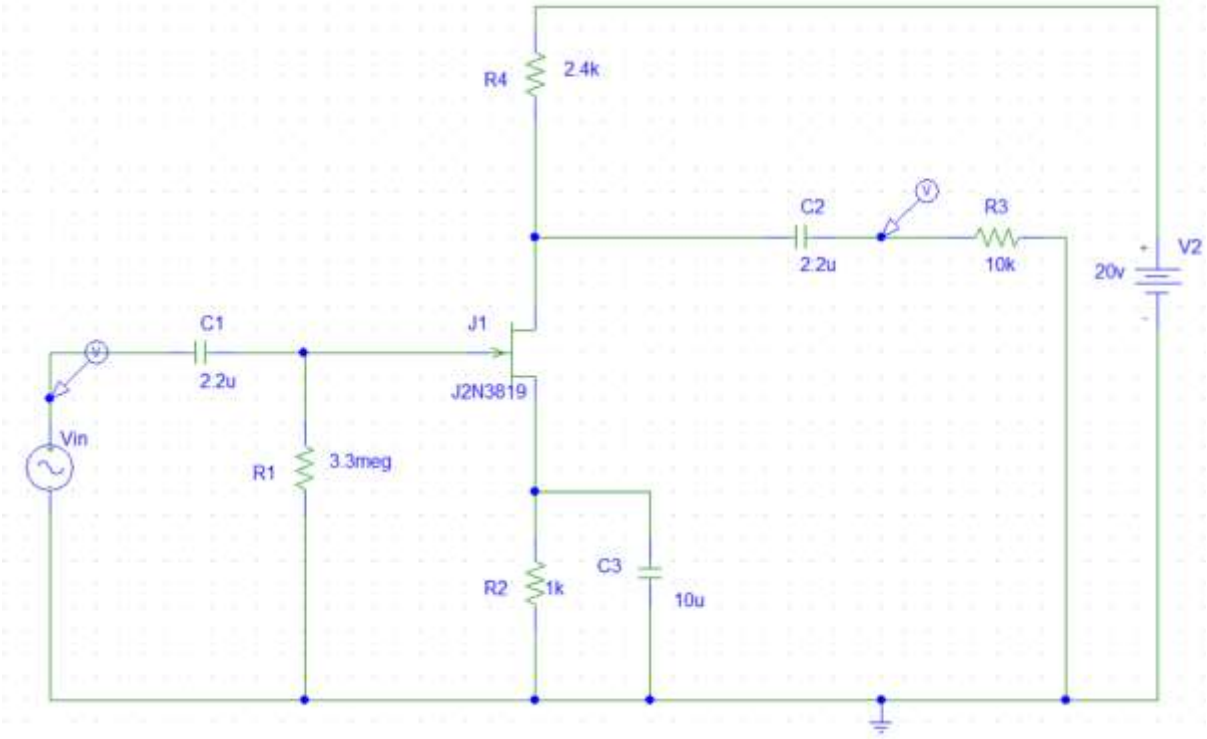



Fig: JFET as an Amplifier Using Pspice

**PROCEDURE :**

**1. Schematic**

1. Place the N-channel JFET J2N3819 along with resistors $R_1, R_2, R_3, R_4$ and capacitors $C_1, C_2, C_3$ according to the schematic layout.
2. Wire $V_2$ (20 V DC) for DC biasing and $V_{\text{in}}$ (VSIN / VAC) for the AC source signal.
3. Attach voltage probes at the input node (across Vin) and output node (across $R_3$, past $C_2$).

**2. Transient Analysis Setup**

1. Set source $V_{\text{in}}$ parameters: DC=0, AC=0.01V, VOFF $= 0$ V, VAMPL $= 0.01$V, FREQ $= 1$ kHz.
2. Configure **Transient Analysis**:
    - **Print Time:** 0ns
    - **Final Time:** 0.002s
3. Run simulation and trace $V_{\text{in}}(t)$ and $V_{\text{out}}(t)$.

**3. AC Sweep / Frequency Response Setup**

1. Configure **AC Sweep Analysis**:
    - **AC Sweep Type:** Decade
    - **Start Frequency:** 1 Hz
    - **End Frequency:** 1 gig
    - **Points/Decade:** 10
2. Plot magnitude voltage gain in decibels (dB) using trace express: dB($V_{\text{out}}/V_{\text{in}}$).

**OBSERVATION TABLE:**

**1. Transient Analysis**

| Vin | Vout | Av | Phase Shift |
|---|---|---|---|
| | | | |

**2. Frequency Response**

| Vmid | Vc | $F_L$ | $F_H$ | BW=$f_H$ - $f_L$ |
|---|---|---|---|---|
| | | | | |

**CALCULATIONS:**

**1. Transient Analysis**

- **Voltage Gain Calculation:**

$$A_v = \frac{V_{\text{out(peak)}}}{V_{\text{in(peak)}}}$$

**2. Frequency Response Analysis**

- **Maximum Mid-Band Voltage Output ($V_{\text{mid}}$):**
  Flat mid band region of frequency response having maximum output frequency.

  Vmid= _____

- **Cutoff Voltage Level (Vc):**

$$\text{Vc} = \frac{\text{Vmid}}{\sqrt{2}} = \text{_____}$$

- **Lower cutoff frequency ($f_L$):**
  On the rising edge (low-frequency response), locate where $V(C2:2) \approx$ Vc

  $f_L$= _____

- **Upper cutoff frequency ($f_H$) :**
  On the falling edge (high-frequency response), locate where $V(C2:2) \approx$ Vc

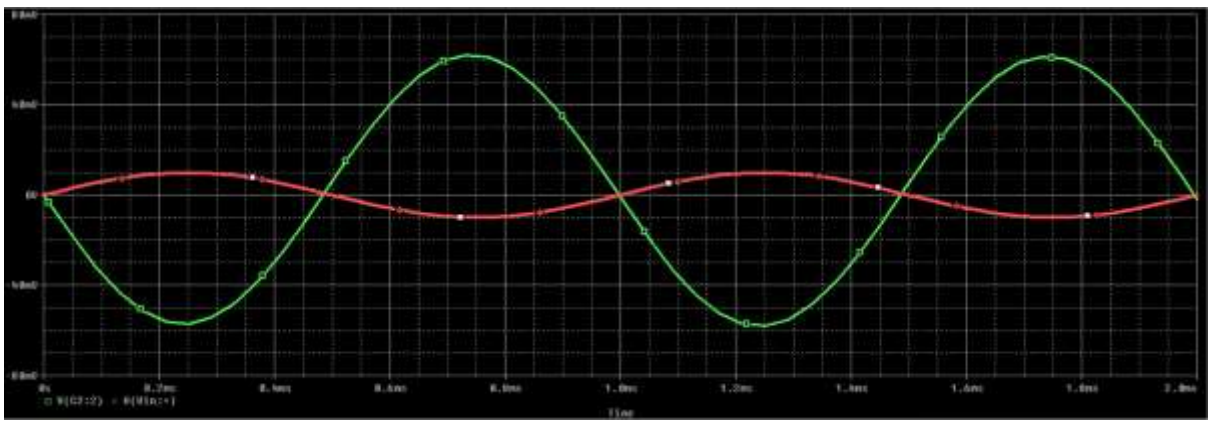
  $f_H$ =_____

- **Bandwidth:** Calculated using:

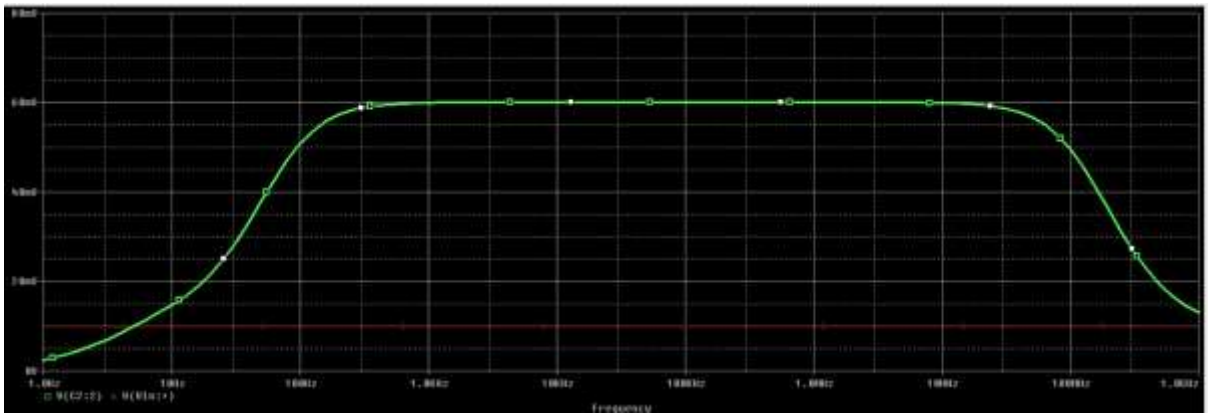
$$\text{BW} = f_H - f_L$$

**OUTPUT:**

**Transient Response:**



**Frequency Response:**



**RESULT:**

**VIVA VOCE:**

1. How does a JFET amplifier compare to a BJT amplifier in terms of frequency response and input impedance?
2. What is Gain-Bandwidth Product (GBW), and why is it constant?
3. What is the Miller Effect, and how does it affect high-frequency response?
4. What are the 3 dB or half-power points?
5. Why is the frequency response graph of a JFET amplifier divided into three regions?